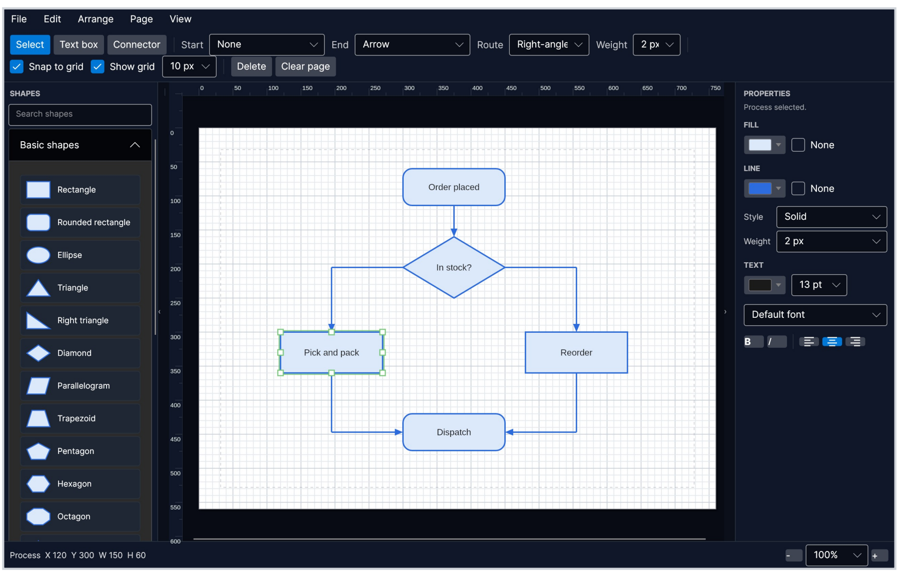
AVASvgMaker

User Guide

A Visio-style diagram editor · version 1.11.0



Contents

Getting started

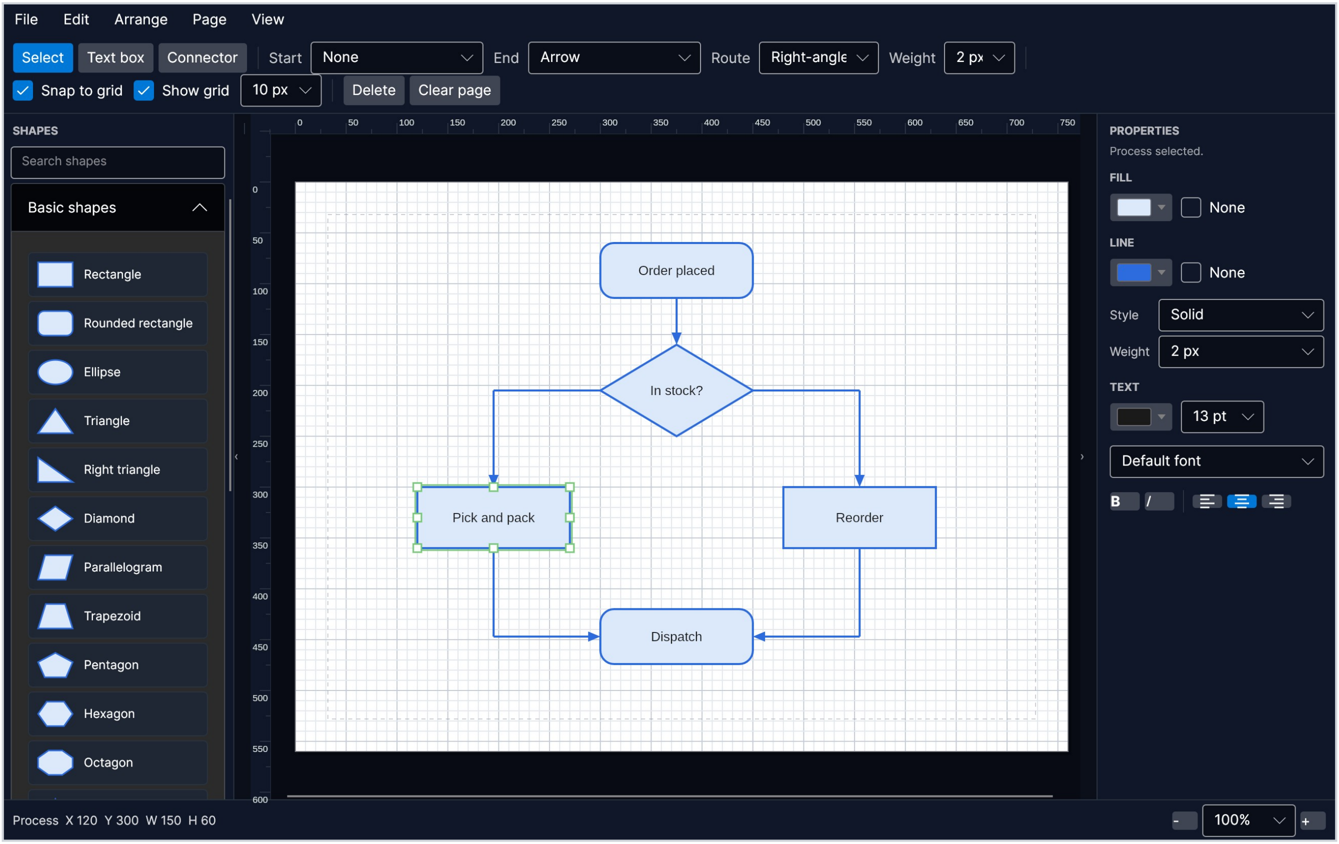
AVASvgMaker is a diagram editor. You drag shapes from the toolbox onto a page, label them, and join them with connectors that route themselves around whatever is in the way.

There is nothing to install. Download the file for your machine from the releases page, unpack it and run it: the .NET runtime and everything else it needs are inside the one file.

- On Linux and macOS the archive keeps the executable bit. If it goes missing on the way, `chmod +x AVASvgMaker`.
- macOS will want the file cleared from quarantine before it will open something downloaded from the web.

Your work is saved in `.avadiag` files. That is the format to keep drawings in - it holds everything, including the things an exported picture cannot. Exports are for handing your drawing to someone else.

The window



Down the left is the **toolbox**, holding every shape you can draw, in categories that fold away, with a search box at the top. Down the right is the **properties panel**, which formats whatever is selected. In the middle is the **page**, floating on a workspace, with **rulers** along the top and left edge and **page tabs** along the bottom once a document has more than one page. The **status bar** at the foot tells you what is selected and what a drag is about to do.

Both side panels fold away when you want the room: `F9` hides the toolbox, `F10` the properties panel, or use the small arrows at their inner edges.

The **toolbar** above the page holds the things you reach for most: the three tools, the ends and routing of connectors, the grid, and Delete.

Drawing shapes

There are two ways to put a shape on the page, and both start in the toolbox:

- **Drag** the shape from the toolbox onto the page.
- **Click** the shape in the toolbox, then click the page where you want it. The toolbox item stays lit until you place it, so it is clear what a click is about to do.

Once a shape is on the page:

To do this	Do that
Move it	Drag it
Resize it	Drag one of the eight handles round it
Turn it	Drag the round handle on the stalk above it
Label it	Double-click it, or select it and press `F2`, then type
Nudge it	The arrow keys, or `Alt` and the arrows
Delete it	Select it and press `Delete`

Positions and sizes snap to the grid, which you can switch off or resize from the toolbar. Nudging moves the selection one grid step at a time, and a container takes its contents along.

The arrow keys nudge whenever the page has the keyboard. After you have used a panel or a toolbar box the arrows belong to whatever you touched last, so **hold** `Alt` and they nudge from anywhere in the window.

Selecting

Click a shape to select it. Hold `Shift` or `Ctrl` and click to add more, or sweep a box across the page to take everything it encloses - the box takes what it fully surrounds, so you can sweep across a large background shape without picking it up.

A selection of several shapes moves, stretches, restyles and deletes as one. Drag one of the handles round it to stretch the whole lot, each shape keeping its place and size in proportion.

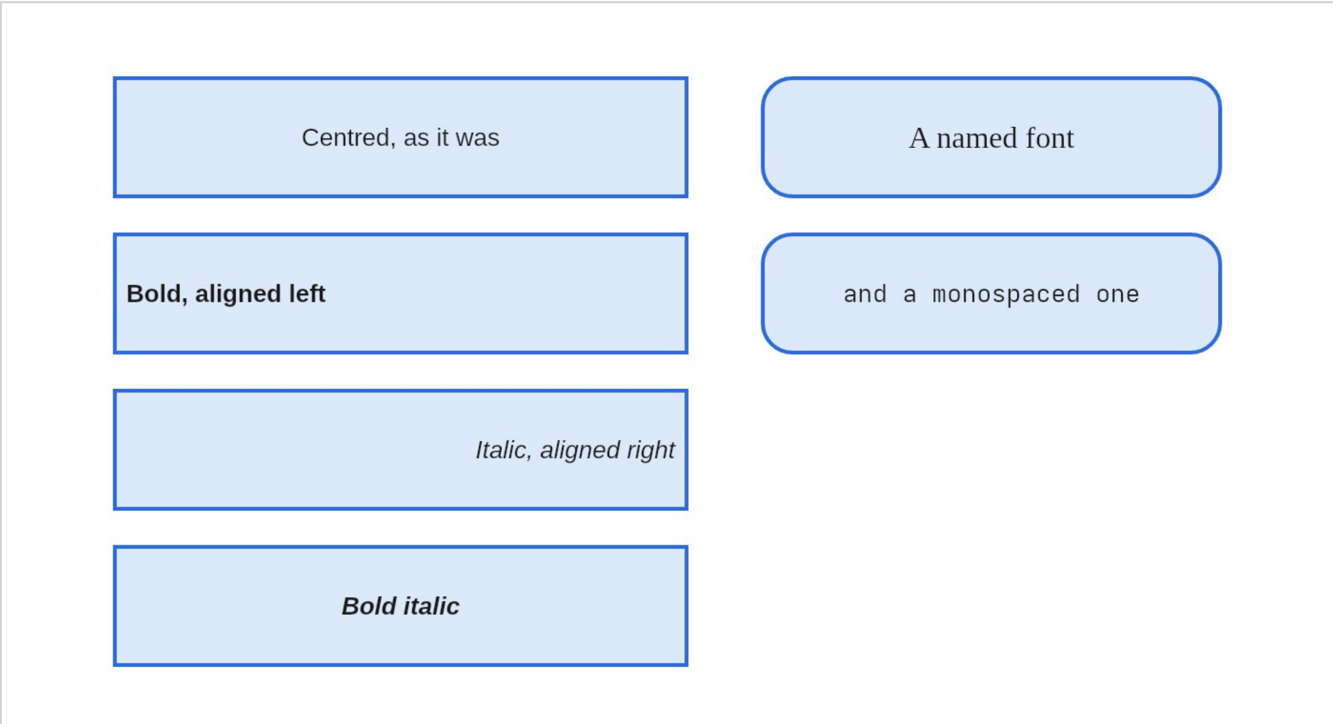
Grouping

`Ctrl+G` makes one thing of two or more shapes. After that, clicking any member picks the whole group, and it moves and resizes together. `Ctrl+Shift+G` breaks it up again, from any member.

Groups do not nest: grouping a group with something else makes one larger group.

Text

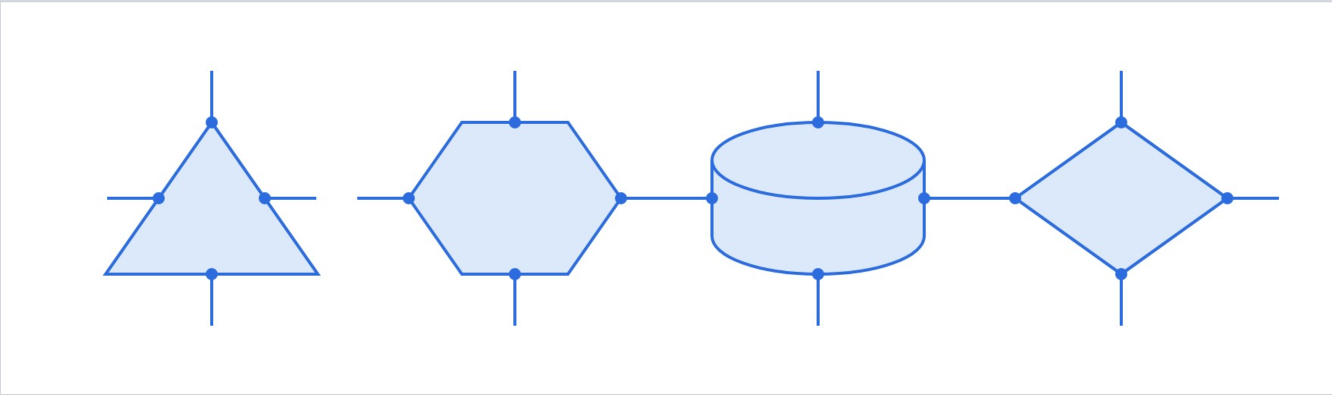
Double-click any shape, or select it and press `F2`, and type its label. Press `Escape` or click away to finish. For text with no shape behind it, use the **Text box** tool in the toolbar and click the page.



The TEXT section of the properties panel sets the colour, the size, the font, bold, italic, and whether the label sits to the left, in the middle or to the right of its shape. The font list is what is actually installed on your machine.

Connecting shapes

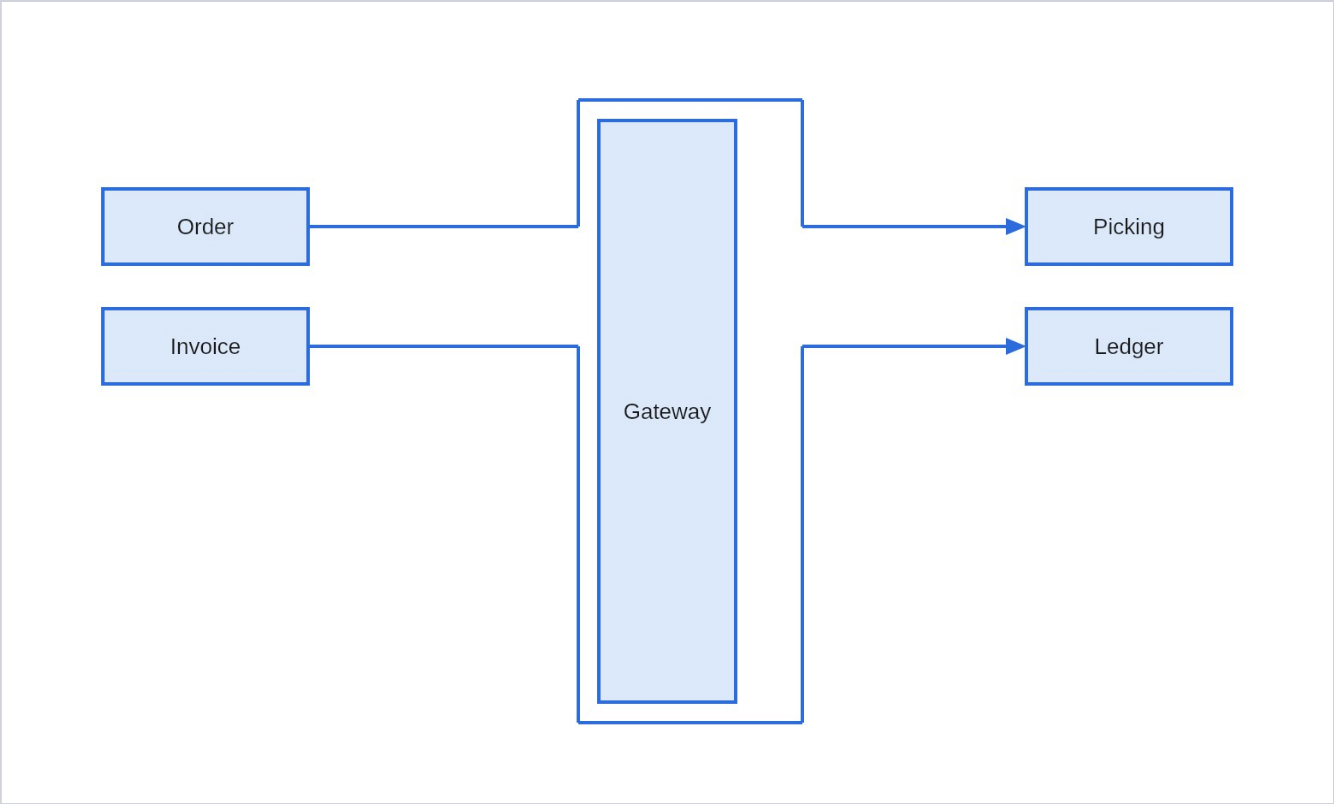
Choose the **Connector** tool in the toolbar and drag from one shape to another. Each end snaps to the nearest connection point, and sticks there: move either shape afterwards and the line follows.



Every shape offers four connection points, and they sit on the shape itself rather than on the box around it - a line meets a triangle on its sloping side and a cylinder on the curve of its cap. They light up while you are drawing a connector.

You do not have to land exactly on the shape. Dropping an end within a grid step of one of its connection points sticks to it just the same, which matters for shapes whose outline sits well inside the box around them - a diamond, an ellipse, a triangle. Drop an end well clear of everything and it stays loose, as it should.

Route in the toolbar chooses between a straight line and a right-angled one. A right-angled connector keeps clear of the shapes in its way, and of the other connectors, and re-routes itself whenever anything it depends on moves.



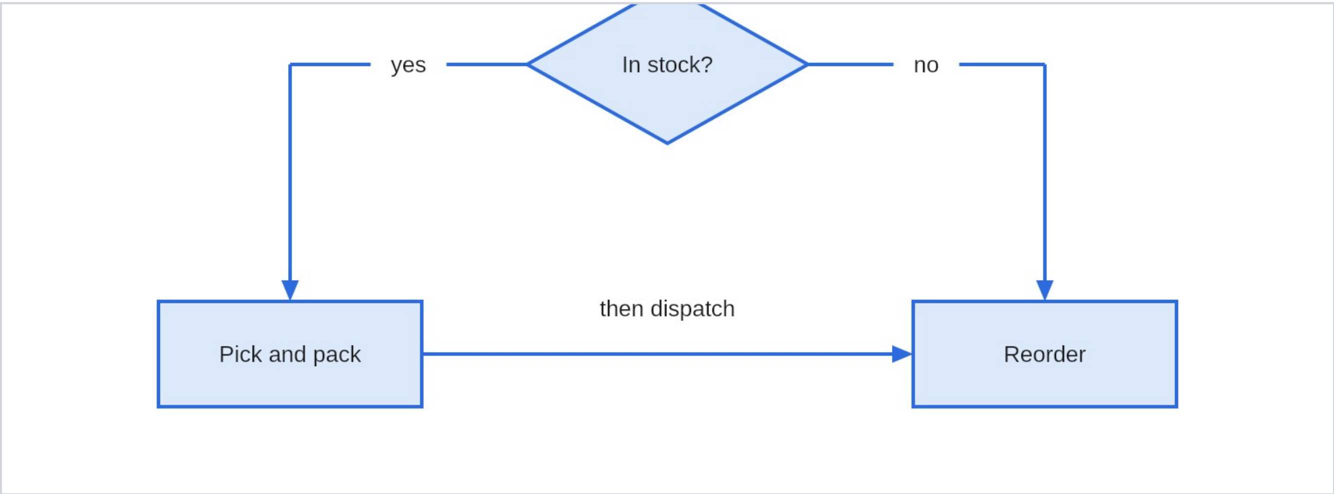
Adjusting a connector by hand

Select a connector and it offers a grab point on each end, on every bend, and in the middle of every straight run:

Handle	What dragging it does
An end	Re-routes that end. Drop it on a shape to stick it there, or on the page to set it free
A bend	Moves that bend. Drop it back on the line between its neighbours and it is removed - the handle turns red first, so you can see it coming
A middle	Slides that run sideways, adding bends either side of it

Double-clicking a bend also removes it. **Edit -> Reset connector route** hands a connector back to the router and forgets everything you placed by hand.

Labelling a connector



Double-click a connector, or select it and press `F2`, and type. The words sit in a gap cut out of the line, so they read cleanly whatever is behind them, and they land in the middle of the longest straight run rather than on a bend.

Drag the label to move it off the line. What is remembered is how far you moved it, so it keeps its place as the shapes move and the line re-routes beneath it. **Edit -> Reset connector route** puts it back.

Line ends

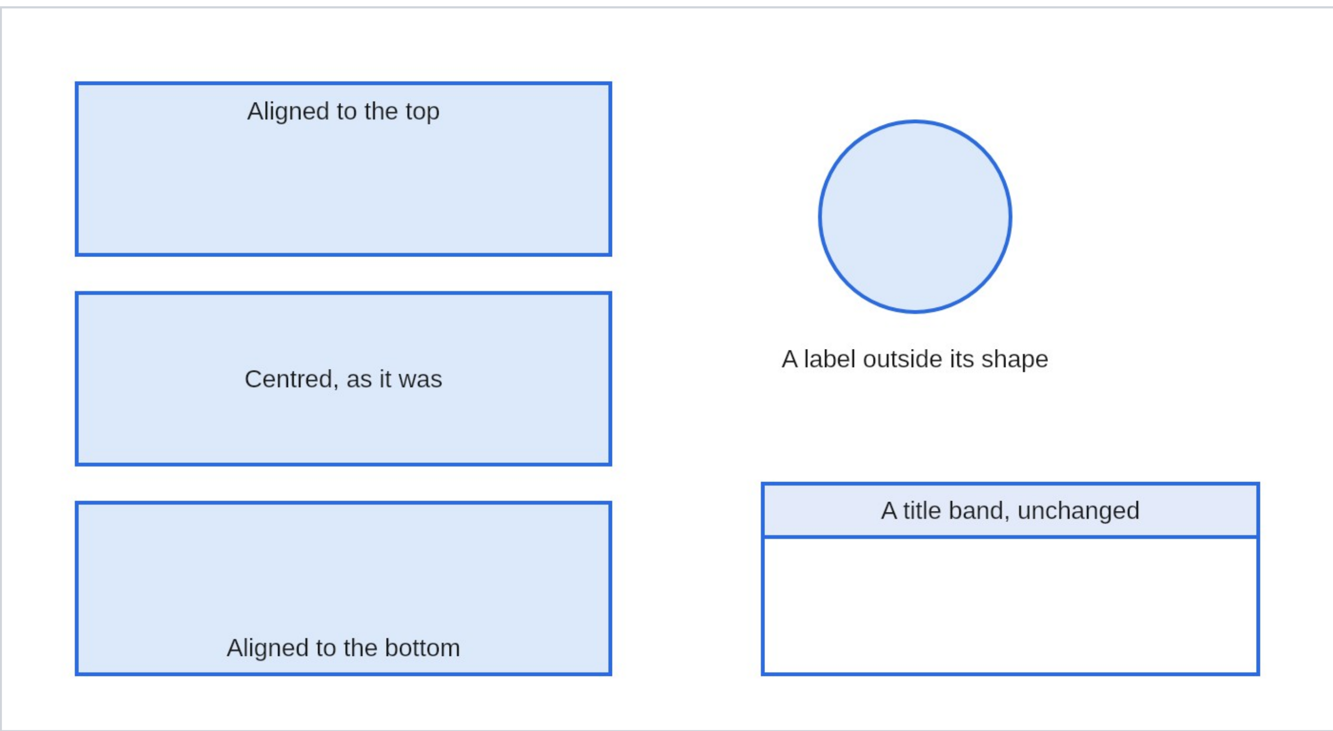
The **Start** and **End** pickers in the toolbar set what each end of a connector looks like: none, an arrow, an open arrow or a dot for ordinary use; a diamond, hollow diamond or hollow arrow for UML; and the crow's foot family for entity-relationship diagrams. What you choose becomes the default for the next connector you draw.

Formatting

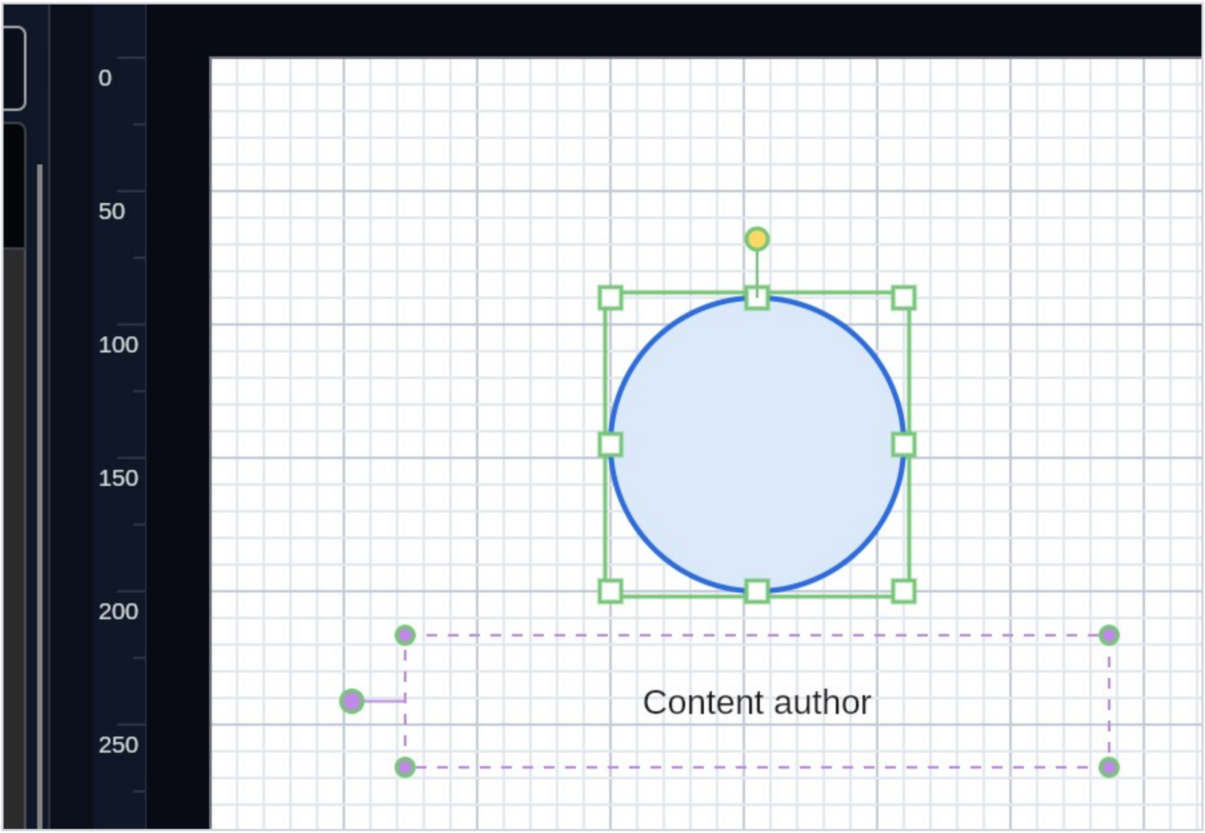
The properties panel down the right formats everything selected at once.

Section	What it sets
FILL	The colour inside a shape, from a palette or a hex value, or None for no fill at all - and a fade to a second colour
LINE	The outline colour or None, the style - solid, dashed or dotted - and the weight
TEXT	The colour, size, font, bold, italic and alignment of the label

There are two rows of alignment buttons: the first puts the label to the left, the centre or the right of its shape, the second puts it at the top, the middle or the bottom.



Moving a label off its shape



A label can also leave its shape altogether - a name under a figure, or a caption beside a box.

Select a shape that has a label and a small round **grip** appears out to its side, on a stalk. Drag it and the label goes with it; a dashed box shows where the words are being wrapped. The shape itself stays where it is - the grip is off to the side precisely so that taking hold of the middle of a shape still moves the shape.

Once a label has a block of its own, the block has **corners**. Drag one to make it wider or taller - which is how a label taken off a narrow shape stops wrapping to that shape's width.

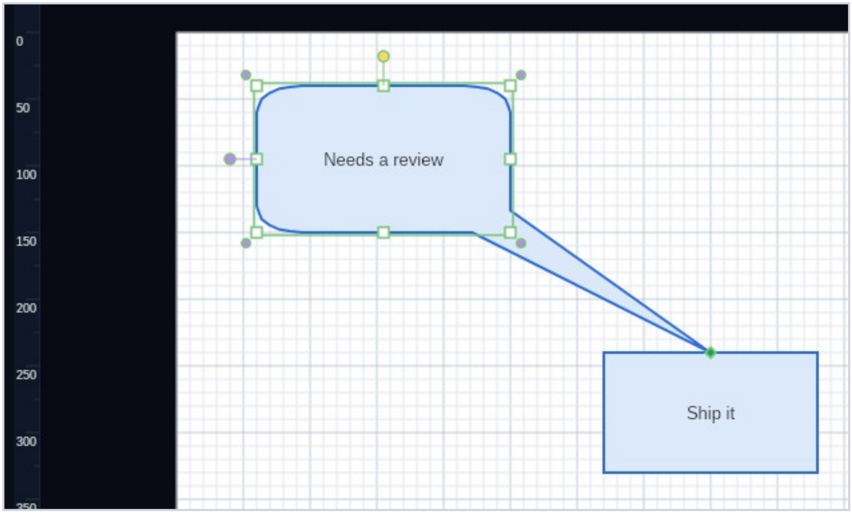
Edit -> Reset label position puts the whole thing back in the middle.

A label that has been moved stays with its shape: move the shape and the label follows, resize it and the label moves and stretches in proportion, turn it and the label swings round with it. It is saved with your drawing and it survives a trip out to Visio and back.

While the block is still the shape, its corners sit a little outside the shape's own so that you can grab either; once the label has been moved they sit on the block itself.

Where the selected shapes disagree, a control shows a dash rather than pretending they match; choosing something then applies it to all of them. With nothing selected the panel sets the formatting for the next shape you draw.

Aiming a callout



A callout is a bubble with a tail, and the tail is the point of it - a callout that cannot be aimed is only a box with a spike on it.

Select a speech bubble, oval callout, rectangular callout or thought bubble and a small **diamond** appears on the tip of its tail. Drag it and the tail follows, out to wherever you want the callout to be pointing. The bubble itself stays where it is and keeps its whole box for the words; only the tail moves.

Drop the tip on another shape's **connection point** and the tail is pinned there. The diamond turns green to say so, and from then on the tail follows that shape: move it across the page, resize it, turn it, and the callout stays aimed at the same point on it. This is the same snap a connector uses, so you do not have to land exactly on the point - near enough will do.

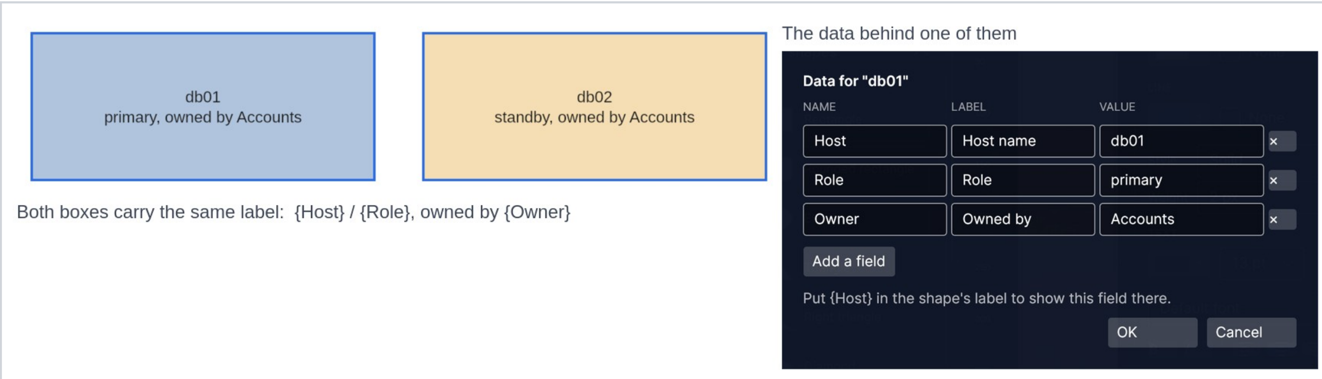
Dropped anywhere else, the tail simply stays where you put it. It is held as a position relative to the bubble, so it travels with the bubble as you move it and stretches with it as you resize it.

Delete the shape a callout was pointing at and the callout is left pointing at the same spot rather than springing back.

A thought bubble has no spike - it trails smaller bubbles instead - but its tail is aimed in exactly the same way, and the trail follows.

Where a callout points is saved with your drawing, and aiming one is a single step to undo.

Data behind a shape



Both boxes carry the same label: {Host} / {Role}, owned by {Owner}

A shape can carry **data**: named fields with values. Select a shape and press `F4`, or use **Edit -> Shape data...**, to see what it carries and to add, change or remove fields.

To put a field on the page, name it in the shape's label between braces. A label of

```
{Host}
{Role}, owned by {Owner}
```

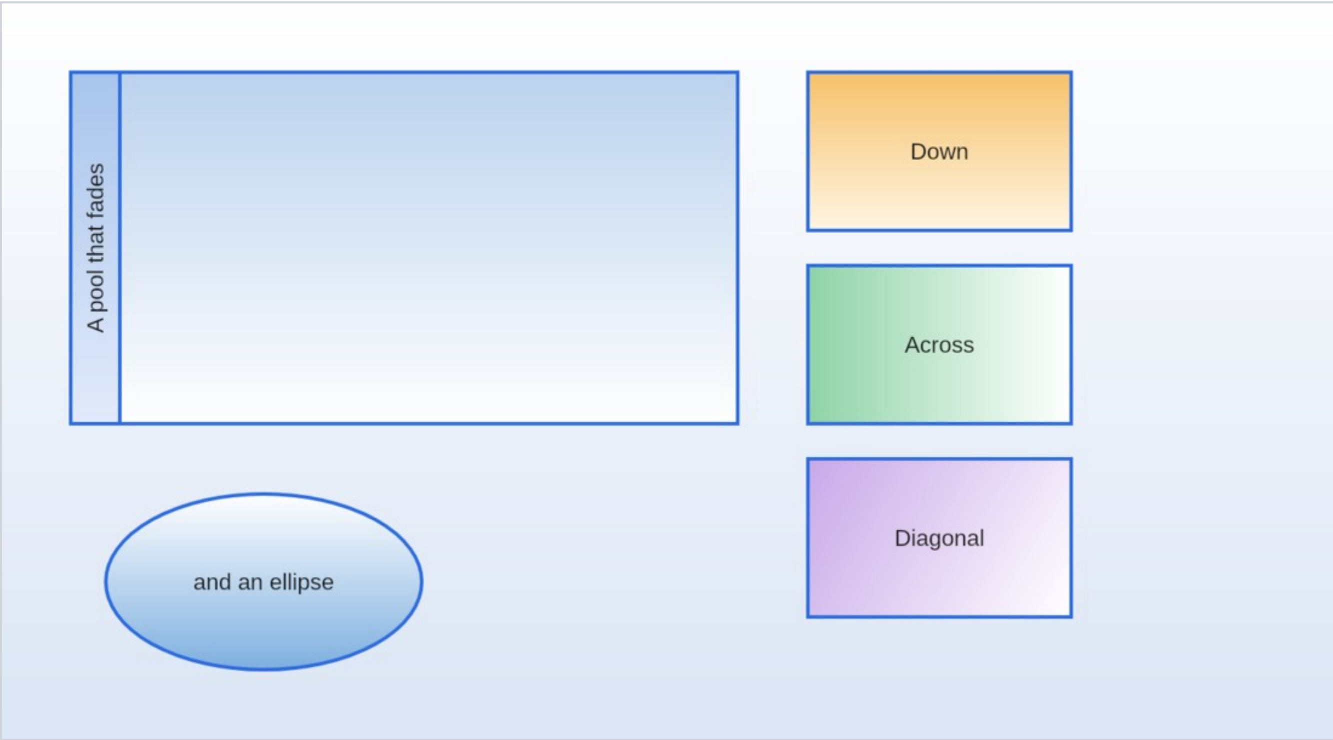
draws as **db01 / primary, owned by Accounts** - and when you change the value, the label changes with it. Both boxes in the picture have exactly that label; all that differs is the data behind them.

Only a name that matches a field is replaced, so a label that happens to contain braces is left alone. Case does not matter: `{host}` finds a field called `Host`.

A field has a **name** - what the label calls it - and can have a **label** of its own for when the name is not what you would want to read, such as `Owner` in the text and "Owned by" in the dialog.

The data belongs to the shape. It is saved with your drawing, and it travels to Visio and back: a Visio drawing whose stencil defines fields arrives with those fields ready to fill in.

Fades



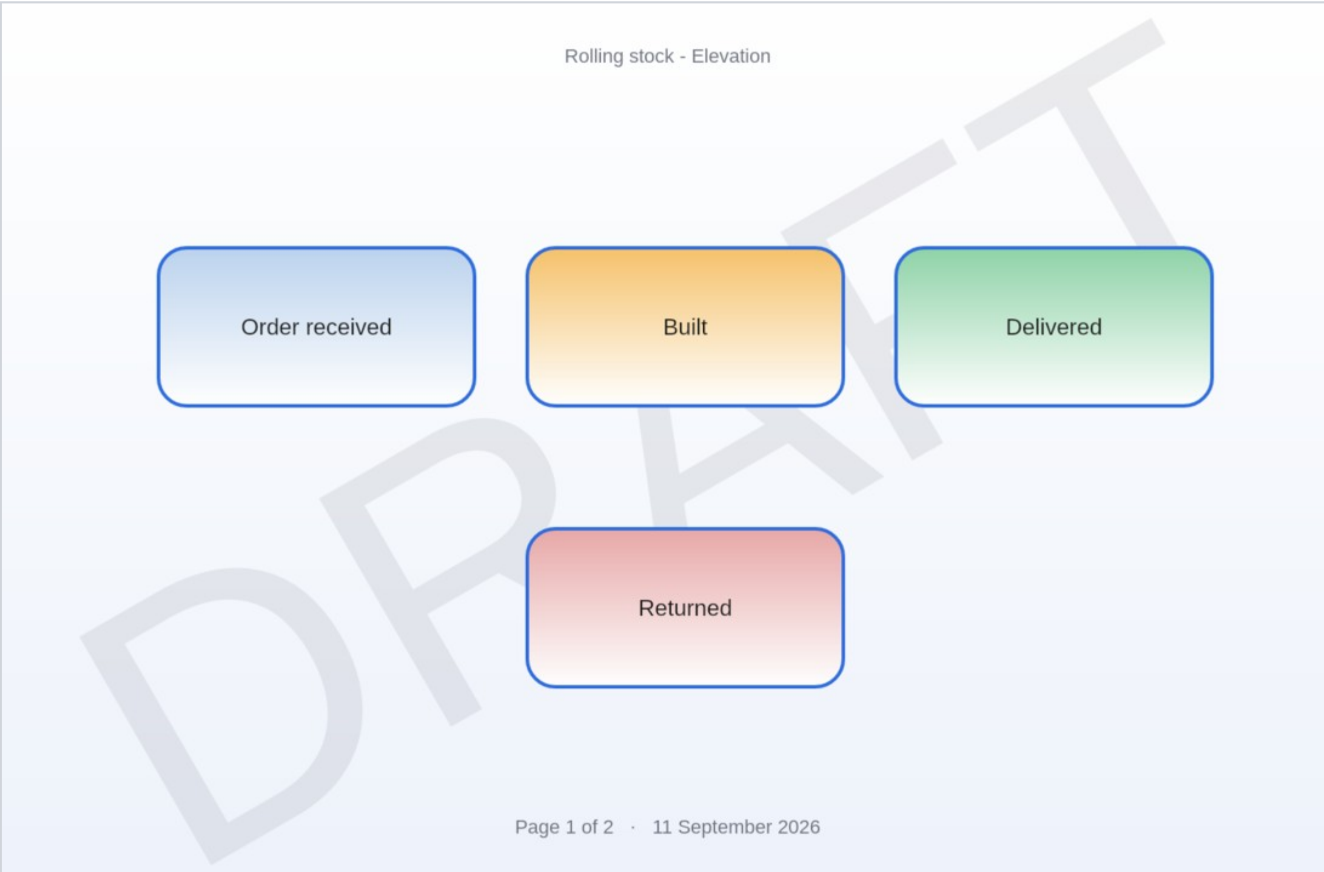
A shape can be filled with a **fade** that runs from one colour to another. Tick **Fade to** in the properties panel, pick the second colour, and choose which way it runs: down, up, across, back or diagonally. The fill colour you already had stays the near end, so turning the fade off again leaves the shape as it was.

It works on pools, lanes and containers too, which is where it is most useful - a band of colour at a lane's heading that thins out across the rest of it.

The **paper** can fade the same way, in **File -> Page setup**.

A shape's fade travels to Visio and back, and so does the paper. A Visio drawing that fades through more than two colours keeps its two ends.

Watermarks, headers and footers



File -> Page furniture... puts words across the page and a line of text along the top and the bottom.

None of it is a shape: you cannot click it, drag it or glue a connector to it. It belongs to the page, so it stays where it is put.

A header or footer can be written with these in braces, and they are filled in as the page is drawn:

`{page}`	Which page this is
`{pages}`	How many pages there are
`{name}`	The page's name, as shown on its tab
`{date}` `{time}`	When it was drawn

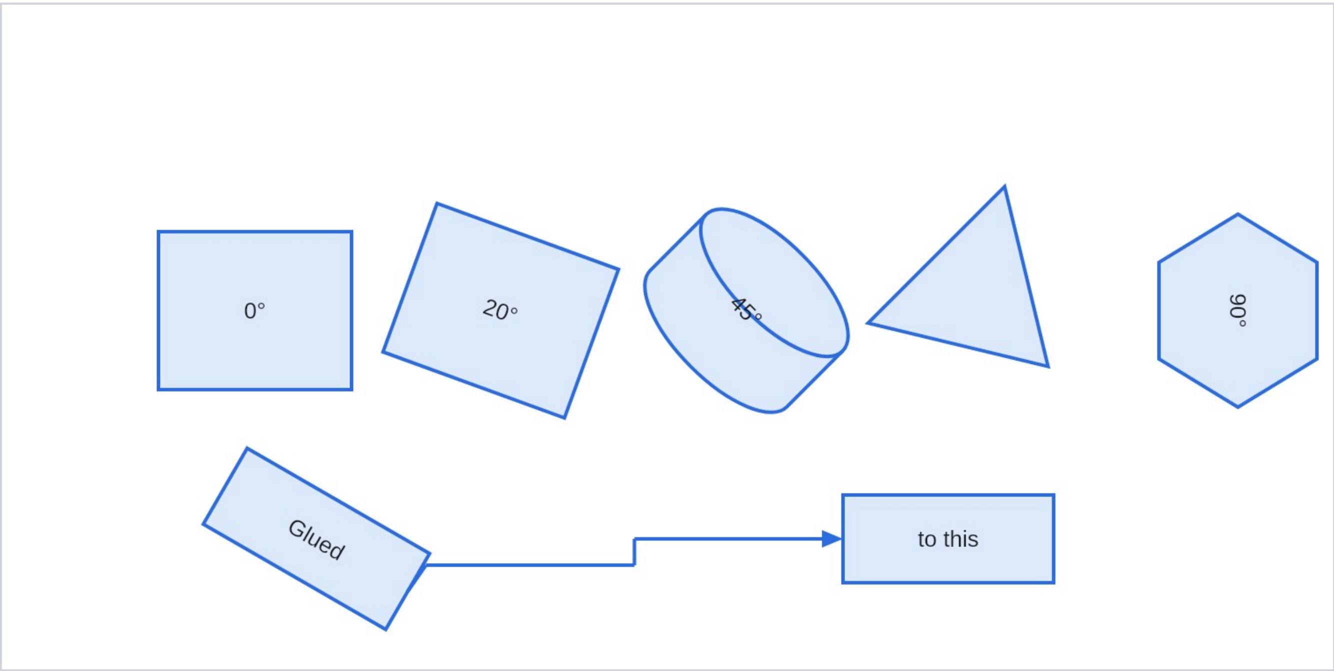
A footer of `Page {page} of {pages}` is therefore right on every page, and stays right when you add, delete or reorder pages.

The watermark is drawn under your diagram so the diagram stays readable, and it is sized to cross the page rather than set in points. Either the watermark or the header and footer can be set on this page alone or on every page at once.

Arranging

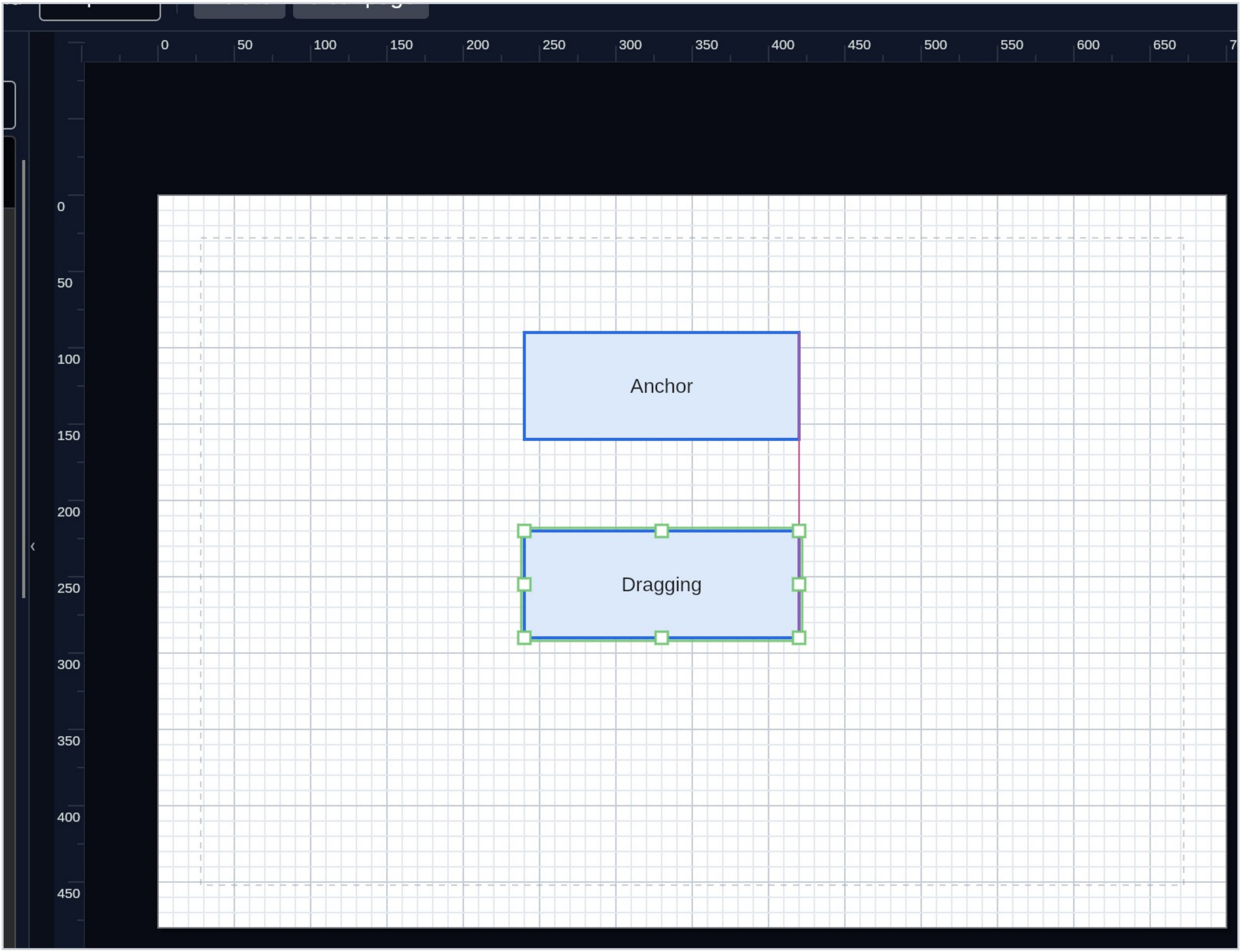
The **Arrange** menu works on a selection:

- **Align** lines shapes up on any of six edges, and **Distribute** spaces them evenly.
- **Make same size** sizes them all to whichever was selected last.
- **Bring to front**, **Bring forward**, **Send backward** and **Send to back** change what is drawn over what.
- **Rotate left**, **Rotate right** and **Straighten** turn shapes in right angles.



For any angle, drag the round handle above a selected shape. Hold `Shift` while you drag to snap to 15°.

Guides and rulers

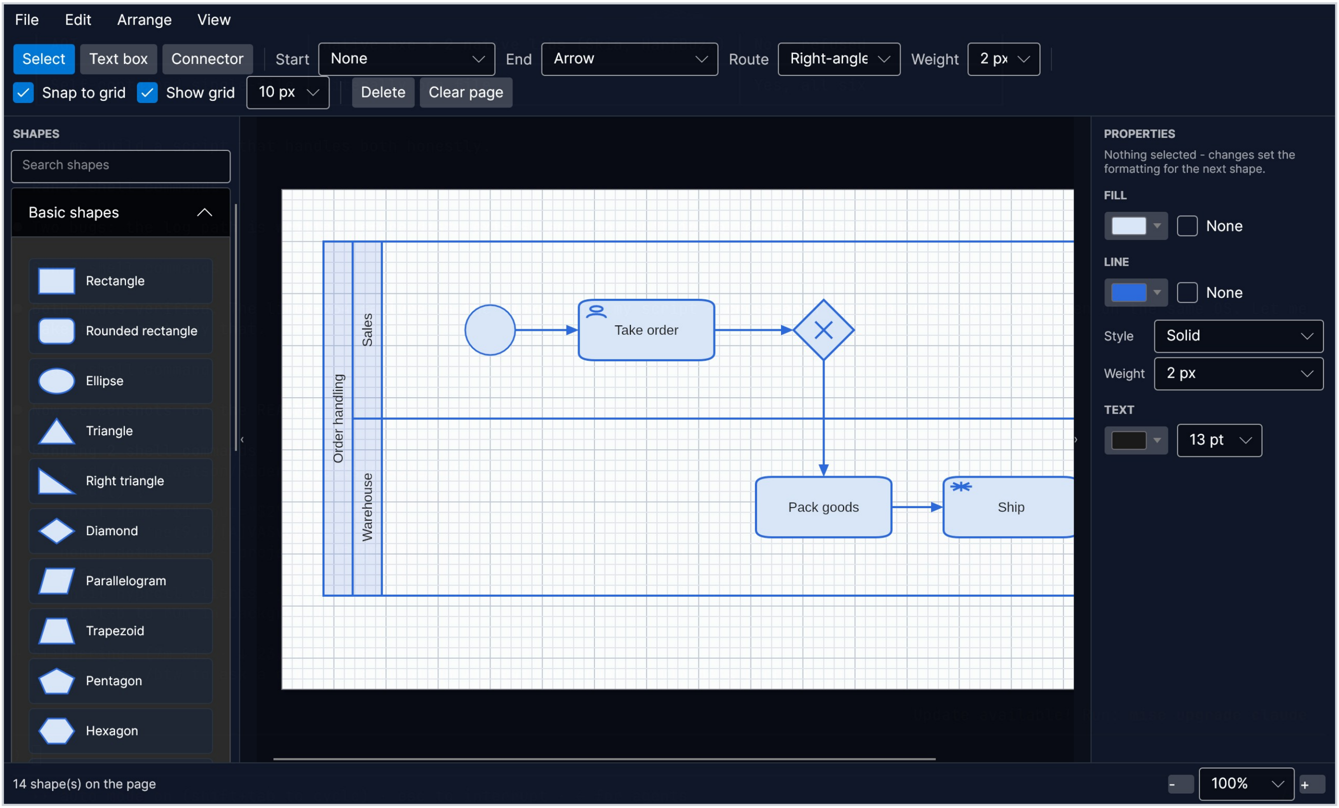


Three things help you line work up, and none of them constrains where a shape may go:

- **Rulers** run along the top and left, marked in page units, with a marker on each following your pointer. `View -> Rulers` folds them away.
- **Smart guides** line a dragged shape up with the edges and middles of the others when it comes close, and draw the line it caught on. `View -> Smart guides` turns them off.
- **The margin** is a dashed inset on the page, set in Page setup and off by default.

The **grid** still catches whichever direction nothing else lined up.

Containers and swimlanes

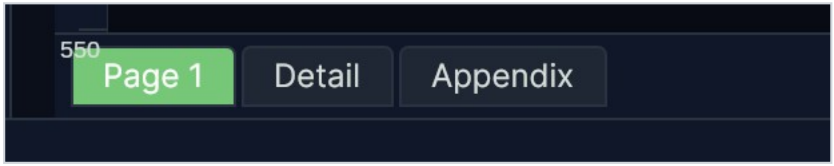


A pool, a lane or a grouping box holds what you put into it:

- **Dropping a shape inside adopts it**, and dragging it out lets it go. The innermost container wins, so a shape dropped on a lane joins the lane rather than the pool around it.
- **Moving a container carries its contents.**
- **Lanes belong to their pool.** They follow when it is moved or resized, and dragging a lane reorders it among its siblings rather than moving it freely.
- **Drag the line between two lanes** to change their heights: one gains what the other gives up. **Arrange -> Even lane heights** puts them back on equal shares.
- **Deleting a container asks** whether to take the contents with it, or drop them on the page.

Only a container's title band and its border are clickable, so you can sweep a selection box across its middle and drop things into it.

Pages

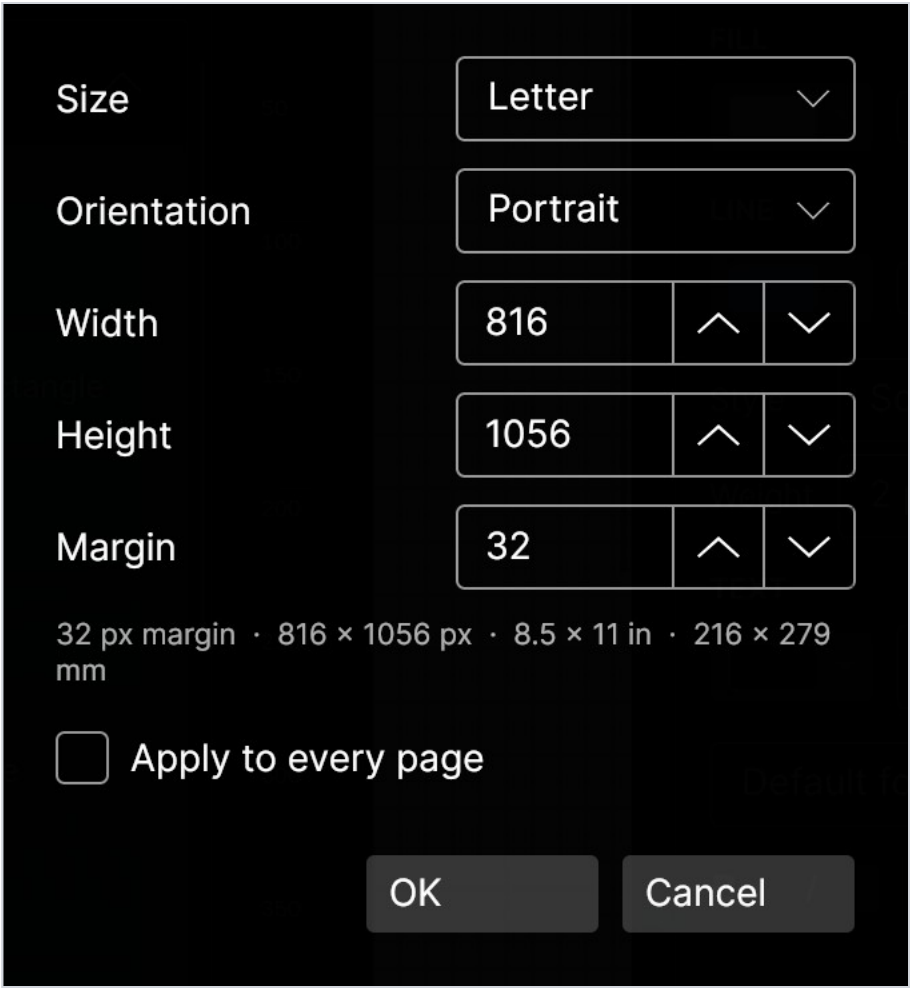


A document holds as many pages as you like. They appear as tabs along the bottom of the drawing area once there is more than one.

To do this	Do that
Turn to a page	Click its tab, or `Ctrl+PageUp` and `Ctrl+PageDown`
Add a page	The `+` at the end of the tabs, or `Ctrl+Shift+P`
Rename one	Double-click its tab
Reorder	Drag the tab along the strip
Duplicate or delete	Right-click the tab

Each page carries its own paper size, so one document can hold a portrait page, a landscape one and an A5 note. Deleting a page with anything on it asks first.

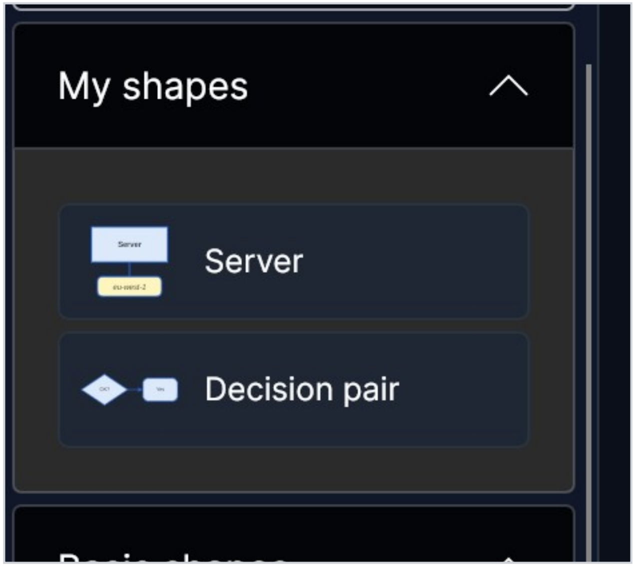
Page setup



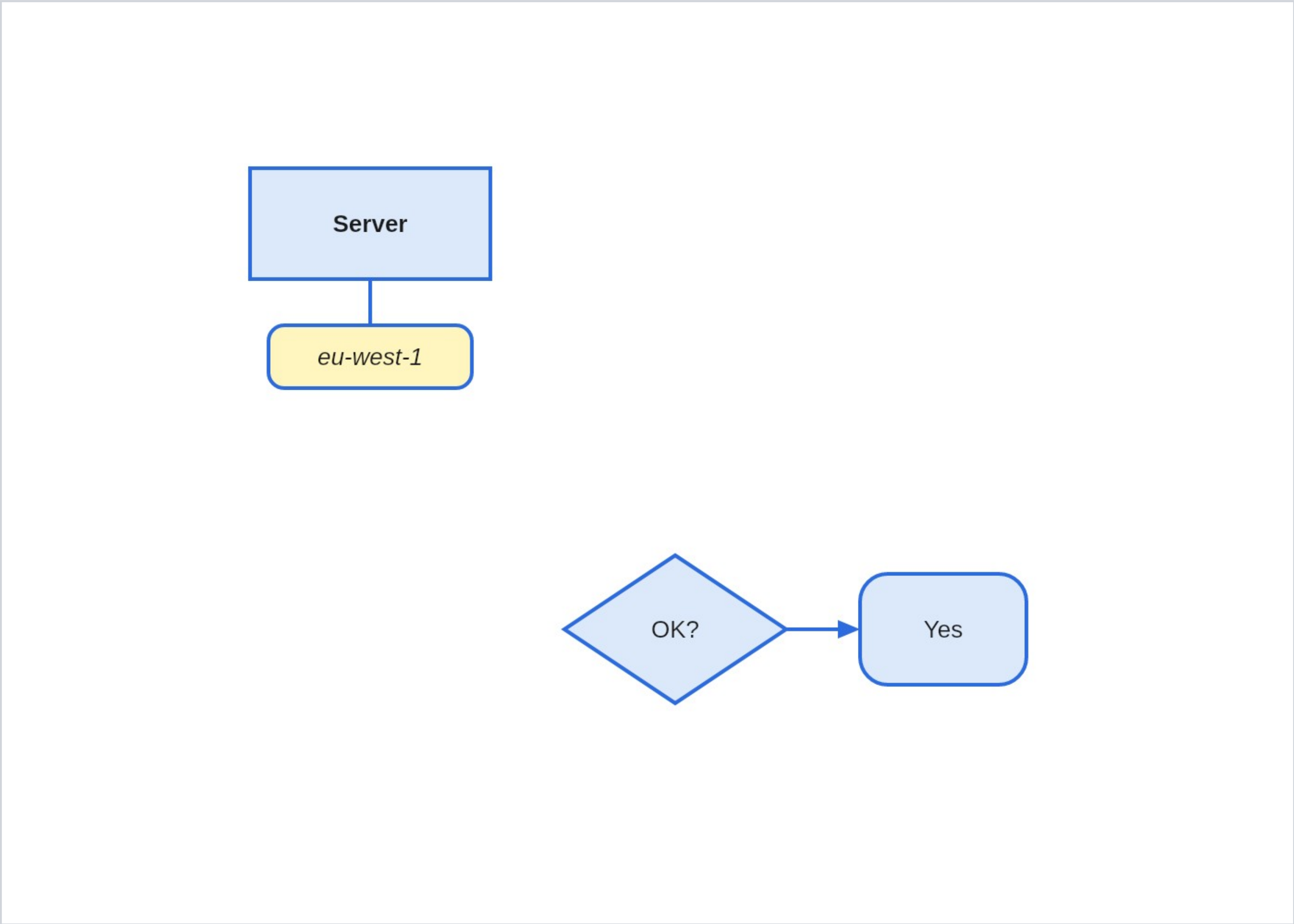
File -> Page set up sets the paper the current page sits on, or every page at once. Pick one of the six presets - Letter, Legal, Tabloid, A3, A4 or A5 - or type a width and height. **Fit to the drawing** sizes the page to what is on it.

Margin draws a dashed inset to line work up against. It is a guide only: nothing is stopped from being placed outside it, and nothing is clipped by it. Set it to 0 for none.

Shapes of your own



Edit -> Save selection as a shape keeps whatever is selected under a name of your choosing. It appears in **My shapes** at the top of the toolbox, and you place it like any other shape. Right-click one to rename or delete it.



What comes back is what you saved: every shape, its colours, its font, the connectors between them and their grouping. Group the selection before saving it if you want it to move as one thing afterwards.

Saved shapes are kept with your settings rather than with a drawing, so they are there in every document and every time you open the app.

Saving and exporting

Save and **Open** use `.avadiag` files. That is the format to keep your work in: it holds the pages, the glue between connectors and shapes, the grouping, the drawing order and every colour and size you chose. Older files always open in newer versions.

File -> Export writes your drawing in seven other formats:

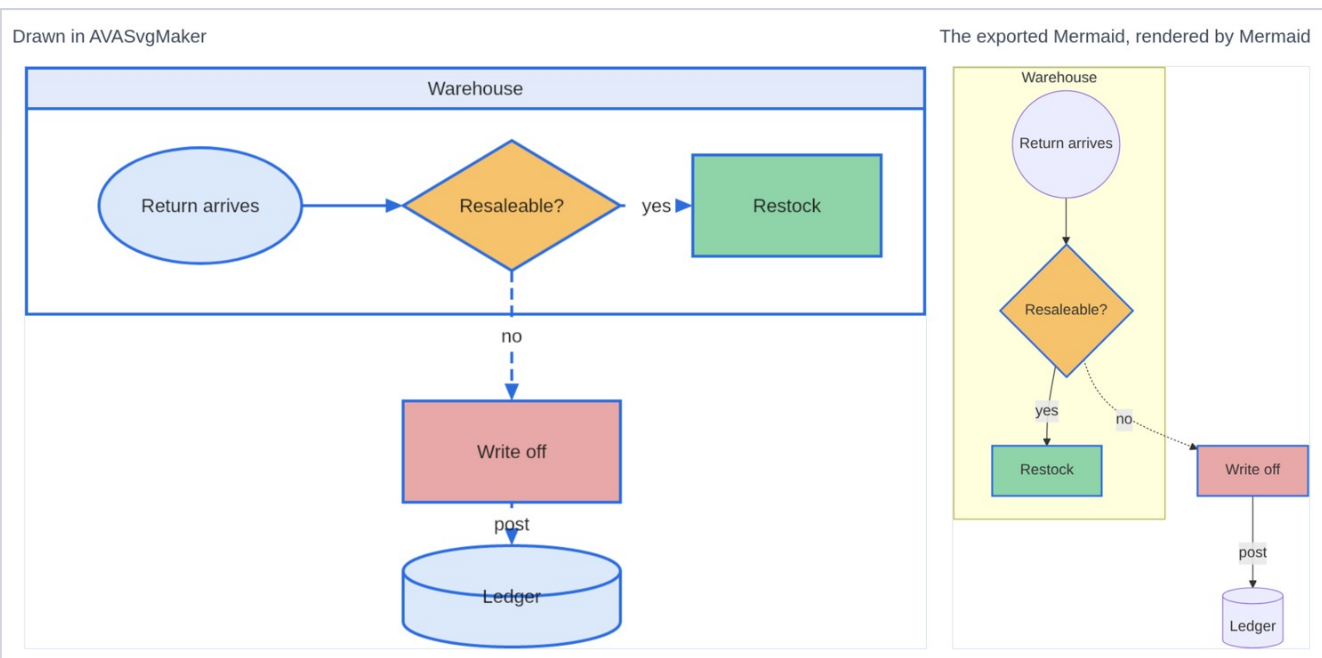
Format	When to use it
SVG	Vector, for handing to another drawing tool
PDF	Vector, at the true physical page size, with the text still selectable. The one to print or attach. Offers all pages or just the one you are on
Visio	A modern `.vsdx` drawing, every page of the document in the one file
Mermaid	The page as a Mermaid `flowchart`, shown on screen to copy rather than saved
PNG	A picture, and the right answer nearly always. Choose 1x to 4x
JPEG	For where nothing else is accepted. A diagram is the worst case for it, so the quality default is high
WebP	Smaller than PNG at moderate quality
BMP	For tools that will take nothing else. A large file for the same picture

Whichever you choose, what is exported is the page and only the page: white paper, no grid, no selection handles and no workspace around it, whatever the screen happens to be showing.

Mermaid

File -> Export -> Mermaid... turns the page into a Mermaid `flowchart` and shows it, coloured and numbered, so you can copy it into a README, a wiki page or a pull request.

Mermaid works out the arrangement itself - there is no way to tell it where things go - so what travels is what your diagram *means*: the boxes, their shapes, their words, what joins what, and which boxes sit inside which container.



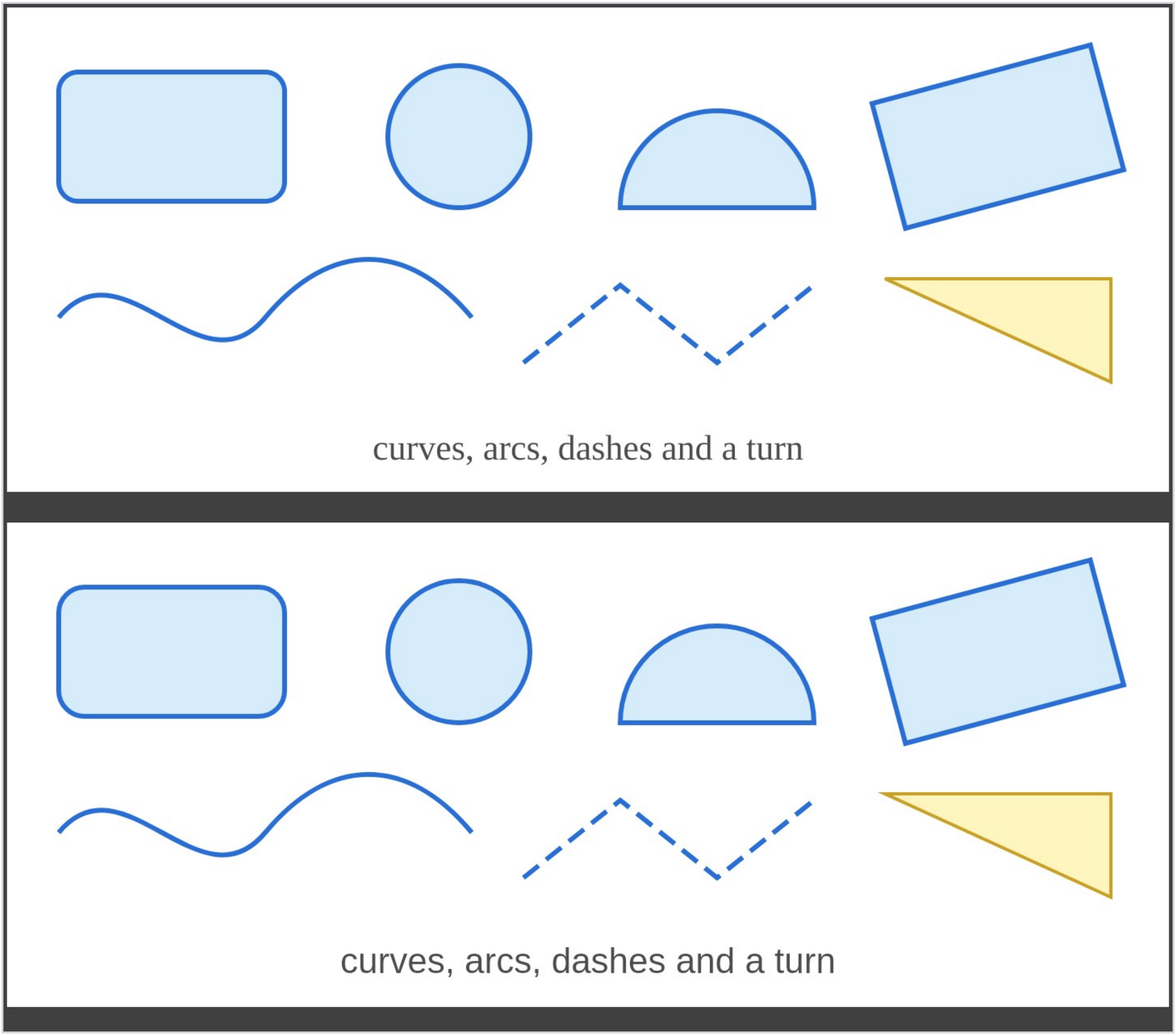
Those are the same diagram. Every shape kept its kind, its colour and its words, the dashed line stayed dashed, and the container came through as a labelled box. What changed is where things are: the three shapes drawn in a row across the top come out stacked, because Mermaid places each one by how far along the chain of arrows it sits.

So a diagram that is a flow of steps travels well. One whose point is its arrangement - a floor plan, a rack layout, a network drawn to match the building - does not, and SVG or PDF is what you want for that.

Anything Mermaid has no room for - a turned shape, a fade, a loose text box, a page's watermark, a connector that is not joined to shapes at both ends - is named in the message rather than quietly dropped.

That last one is worth reading. A connector reported as joined to nothing was never really stuck to the shapes it appears to touch, and would have come adrift as soon as one of them moved - so the message is a way of checking a drawing as well as exporting it.

Importing SVG

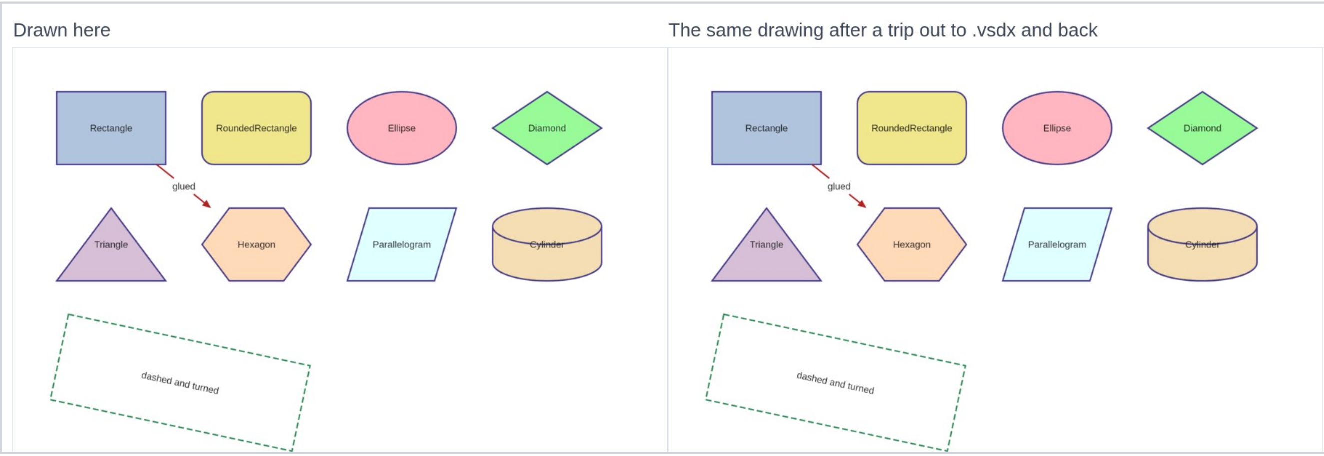


File -> Import -> SVG... reads a drawing in as editable shapes, onto a page of its own so it does not land on top of work you have already done.

Reading SVG is an interpretation, not a copy. SVG can say far more than this editor can hold, so the importer takes what maps onto shapes with a fill, an outline and a label - shapes, paths, curves, arcs, transforms, colours, dashes and text - and **tells you what it could not take**, rather than leaving you to notice.

Gradients, patterns, filters, clipping and embedded images are the usual things it cannot bring in. A drawing that opens with a rectangle the size of itself is painting its background, which here is the page, so that is left behind too.

Visio drawings



File -> Import -> Visio drawing... opens a modern `.vsdx`. Each page of the drawing arrives as a page of its own here, so nothing lands on top of work you have already done.

Most shapes in a real Visio drawing carry no picture of their own: they say which *master* they are a stamp of, and the master holds the geometry. The importer follows that all the way down, including the common case of a shape that redraws only part of what it inherits, so what you get is the drawing as Visio shows it rather than a page of empty boxes.

It reads pages and their sizes, shape outlines - straight edges, arcs, ellipses, curves and splines - fills, line colours, weights and dash patterns, groups along with any turn or flip the group itself has, connectors and which shapes each end is stuck to, and text with its size, weight, slant, colour and alignment. It also reads the block the text sits in, which Visio places separately and which need not be on the shape at all - so the name under a stick figure arrives under the figure.

Whether a connector has an arrowhead on each end is read, though not which of Visio's forty-odd line ends it is - those and the twelve here are different sets, so an end that is drawn comes in as a plain arrow. A connector the drawing gives no arrowhead comes in without one.

Colours come through even when the drawing does not name them. A shape may leave its formatting to a *style* - a named set the drawing keeps to one side, which may itself defer to another - or to the current theme, and both are followed. Most of what you see in a Visio drawing is set this way rather than on the shapes themselves.

Two things it cannot bring in:

- **A themed fill.** A themed *line* comes through. A themed *fill* is tinted by Visio in a way this does not read, so rather than paint the shape solid in its line colour it is left unfilled. A fill the drawing states outright comes in as it is.
- **Gradients, shadows, embedded pictures, layers and shape data.** Nothing here can hold them.

As with SVG, it **tells you what it left out** rather than leaving you to notice.

File -> Export -> Visio... writes one, every page of the document in the one file. Colours, line styles, rotation, text and connector glue all go with it. Everything Visio would normally look up in a stencil is written into the file itself, so it opens without needing one.

One thing worth knowing: the exporter was built to the published file format and its output is read back and checked by this app, but it has not been opened in Visio itself - there is no copy of Visio to try it with. If you have one, the result either way is worth reporting.

Keyboard

`Ctrl+N` `Ctrl+O` `Ctrl+S` `Ctrl+Shift+S`	New, Open, Save, Save As
`Ctrl+E` `Ctrl+Shift+E`	Export SVG, Export PNG
`Ctrl+Z` `Ctrl+Y`	Undo, Redo
`Ctrl+X` `Ctrl+C` `Ctrl+V` `Ctrl+D`	Cut, Copy, Paste, Duplicate
`Ctrl+A`	Select all
`F4`	The data the selected shape carries
`Ctrl+G` `Ctrl+Shift+G`	Group, Ungroup
`Ctrl+Shift+F` `Ctrl+Shift+B`	Bring to front, Send to back
`Ctrl+]` `Ctrl+[`	Bring forward, Send backward
`Ctrl+Shift+P`	Add a page
`Ctrl+PageUp` `Ctrl+PageDown`	Previous page, Next page
`Ctrl+wheel`	Zoom about the pointer
`Ctrl++` `Ctrl+-` `Ctrl+0` `Ctrl+9`	Zoom in, out, actual size, fit the page
`F2`	Edit the label of the selected shape
`F9` `F10`	Fold the toolbox and properties panels away
Arrow keys	Nudge the selection by one grid cell
`Alt` + arrows	Nudge from anywhere in the window
`Delete`	Delete the selection, and any connectors glued to it
`Escape`	Back to the select tool

Middle-drag, or space and drag	Pan the page

Fitting in

On Omarchy the app takes its colours from the current desktop theme and re-colours the moment you switch, falling back to the palette it ships with everywhere else. The page itself always stays white: it is paper, and it is what your exports land on.

If you change the display scale while the app is open it will say so in the status bar rather than half-applying it. Restart it to draw at the new scale.